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CPLN 6710: Statistical and Data Mining Methods for Urban Data Analysis

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K-Means Clustering

Introduction

This assignment uses k-means clustering to identify neighborhood types across 1,720 census block groups in Philadelphia. The analysis is based on five socioeconomic and housing variables: median house value (MEDHVAL), median household income (MEDHHINC), percent of individuals with at least a bachelor's degree (PCTBACHMOR), percent of single or detached housing units (PCTSINGLES), and percent of vacant housing units (PCTVACANT). Together, these variables describe differences in housing value, income, education, housing form, and vacancy across Philadelphia block groups.

The purpose of this analysis is to use k-means clustering to group block groups with similar characteristics. Instead of starting with existing neighborhood boundaries, k-means allows the data itself to identify groups of block groups that share similar socioeconomic and housing profiles. This can help answer questions such as whether Philadelphia block groups fall into distinct neighborhood types, what characteristics define those types, and whether those cluster memberships are spatially patterned across the city.

Before running the cluster analysis, the five variables were standardized using the `scale()` command in R. This step is important because the variables are measured on different scales. For example, median home value and median household income are measured in dollars, while the other variables are percentages. Standardizing the variables prevents variables with larger numerical ranges from dominating the clustering result.

Methods

K-means is a clustering algorithm used for numeric variables, especially when working with larger datasets. It divides observations into a selected number of non-overlapping groups, meaning that each observation belongs to only one cluster. This method is commonly used in data mining and machine learning because it helps identify patterns or groupings in data without requiring predefined categories. Before running the algorithm, we must decide how many clusters, or k , should be created.

The k-means process works through several repeated steps. First, the algorithm selects k initial cluster centers, also called centroids. Next, it calculates the distance between each observation and each centroid. Each observation is then assigned to the cluster with the nearest

centroid. After all observations are assigned, the algorithm recalculates the centroid of each cluster based on the observations currently assigned to it. The distances are then recalculated, and observations may be reassigned to different clusters if they are closer to a new centroid. This process continues until the cluster assignments stop changing or change only very little.

The main goal of k-means is to minimize the within-cluster sum of squared errors, or SSE. This means the algorithm tries to make observations within the same cluster as similar as possible by reducing the squared distance between each observation and its cluster centroid. The variables used in k-means do not have to be latitude and longitude coordinates. However, they should generally be numeric and continuous, or at least represented in a numeric format that makes distance comparisons meaningful.

In this assignment, we standardized the five variables using the `scale()` command in R before running the k-means model. This made the variables comparable even though they were measured on different scales. We then applied k-means using Euclidean distance and multiple random starts to reduce the chance that the final result would be based on a poor initial clustering solution.

K-means has several important limitations. First, we must choose the number of clusters before running the final model. This can be challenging because different values of k may produce different interpretations of the data. Although tools such as scree plots and the `NbClust` package can help guide this choice, the final decision still requires judgment. Second, k-means is designed for numeric variables, so it is not well-suited for categorical variables unless they are transformed into a usable numeric format, such as binary indicator variables. Third, the algorithm works best when clusters are relatively similar in size, density, and shape. If the true groups in the data are uneven, overlapping, or irregularly shaped, k-means may not classify them well. Fourth, k-means is sensitive to noisy data and outliers because extreme values can pull the cluster centers and affect cluster assignments. Finally, because k-means depends partly on its starting points, it may find a local minimum of the within-cluster sum of squared errors rather than the best possible global solution.

In addition to k-means, several other clustering methods could be used. One option is hierarchical clustering, which groups observations step by step based on their similarity. Unlike k-means, hierarchical clustering does not require the researcher to choose the number of clusters before running the algorithm. Instead, the results are displayed in a dendrogram, or cluster tree, and the researcher can decide where to “cut” the tree to define the final clusters. Hierarchical clustering can be useful for smaller datasets because the dendrogram provides a visual way to understand how observations are grouped. However, with a larger dataset such as this one, which includes 1,720 block groups, the dendrogram may become difficult to interpret. For this assignment, k-means is more convenient because it produces a clearer set of neighborhood types.

Another possible method is DBSCAN, or Density-Based Spatial Clustering of Applications with Noise. DBSCAN identifies clusters based on dense groups of nearby observations. Unlike k-means, it can detect irregularly shaped clusters and can classify some observations as outliers rather than forcing every observation into a cluster. DBSCAN depends on two main parameters: *eps*, which defines the search radius around each observation, and *MinPts*, which defines the minimum number of nearby observations needed to form a dense cluster. Points with enough neighbors within the *eps* radius are treated as core points, points near a core point may become border points, and points that do not belong to any dense area are treated as noise or outliers.

A related method is spatio-temporal DBSCAN (ST-DBSCAN). This method extends DBSCAN by considering both spatial distance and time. In simple terms, ST-DBSCAN groups observations that are close to one another in both space and time. It usually requires parameters such as *MinPts*, a spatial distance threshold, and a temporal distance threshold. For this assignment, however, k-means is more appropriate than DBSCAN or ST-DBSCAN. The goal is to classify all Philadelphia block groups based on five socioeconomic and housing variables, not to identify dense point clusters in physical space or across time. Since every block group should receive a cluster assignment and the variables are numeric attributes, k-means is better suited to the research question.

Results

The scree plot is shown in Figure 1, and the NbClust diagnostic results are shown in Figure 2. The scree plot shows a sharp decline in the within-cluster sum of squares when moving from one cluster to two clusters. After that point, the curve continues to decrease, but at a much slower rate. This suggests that the main elbow occurs around $k = 2$, although a solution between two and four clusters could also be considered reasonable based on the shape of the plot. In practical terms, adding more clusters after this point reduces within-cluster variation only gradually.

The NbClust results support the two-cluster solution. In the bar chart, eight criteria selected two clusters, which was the largest number of criteria for any cluster solution. The next strongest result was 15 clusters, selected by six criteria. However, the 15-cluster solution would be much harder to interpret and is less consistent with the main elbow in the scree plot. Because the assignment asks us to use the number of clusters selected by the largest number of NbClust diagnostics, we selected $k = 2$ as the preferred cluster solution.

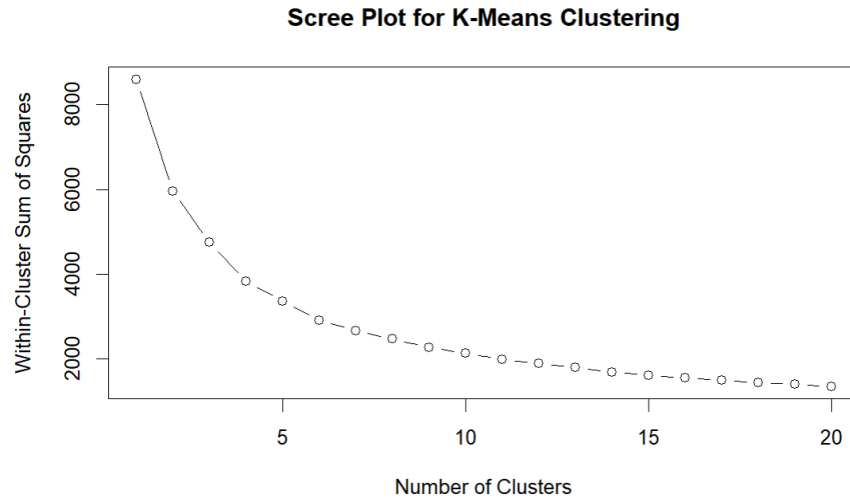


Figure 1: Scree Plot

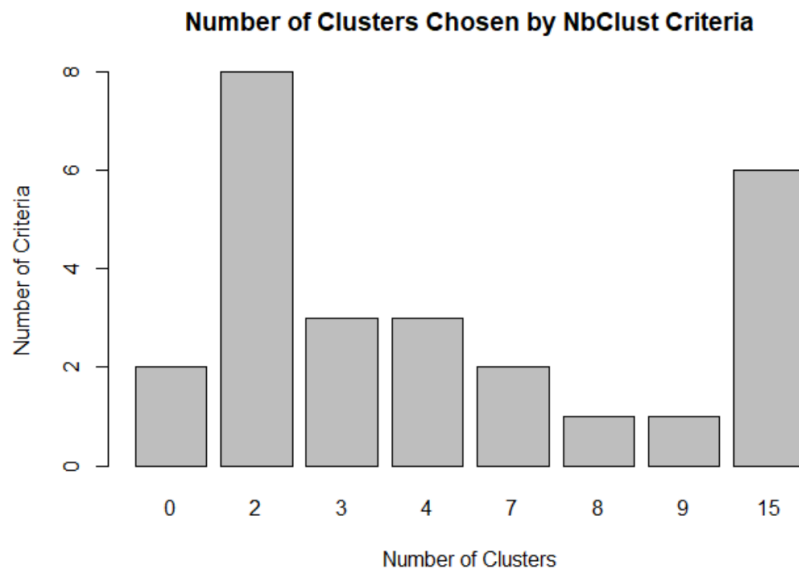


Figure 2: Number of Clusters

The aggregate results are presented in Table 1. The table shows clear differences between the two clusters across all five variables. Cluster 1 has much lower median home values and household incomes, with an average MEDHVAL of about \$49,952 and an average MEDHHINC of about \$27,668. It also has a lower percentage of residents with at least a bachelor's degree, at about 10%, and a lower percentage of single or detached housing units, at about 7%. However, this cluster has a higher vacancy rate, with PCTVACANT around 13%.

In contrast, Cluster 2 has much higher median home values and household incomes, with an average MEDHVAL of about \$152,495 and an average MEDHHINC of about \$51,983. It also

has a much higher share of college-educated residents, at about 47%, and a higher percentage of single or detached housing units, at about 22%. Its vacancy rate is much lower, at about 5%. These patterns suggest that the two-cluster solution is substantively meaningful. Cluster 1 can be described as “Lower-Advantage, Higher-Vacancy Neighborhoods” because it has lower median home values, lower household incomes, lower bachelor’s degree attainment, fewer single or detached housing units, and higher vacancy rates. Cluster 2 can be described as “Higher-Advantage, Lower-Vacancy Residential Areas” because it has much higher home values, higher household incomes, higher bachelor’s degree attainment, more single or detached housing units, and lower vacancy rates.

cluster <dbl>	MEDHVAL <dbl>	MEDHHINC <dbl>	PCTBACHMOR <dbl>	PCTSINGLES <dbl>	PCTVACANT <dbl>	size <dbl>
1	49952.42	27668.47	10.24	6.75	12.52	1446
2	152495.26	51982.58	46.92	22.29	4.82	274

Table 1: Mean Variable Values by K-Means Cluster (k = 2)

Overall, the two-cluster solution makes substantive sense because it separates Philadelphia block groups into two broad socioeconomic and housing profiles. However, the solution should be interpreted as a broad typology rather than a detailed neighborhood classification. Since only two clusters were selected, the model captures the largest divide in the data but may not fully represent more moderate or mixed neighborhood conditions.

Spatial Distribution of K-Means Clusters

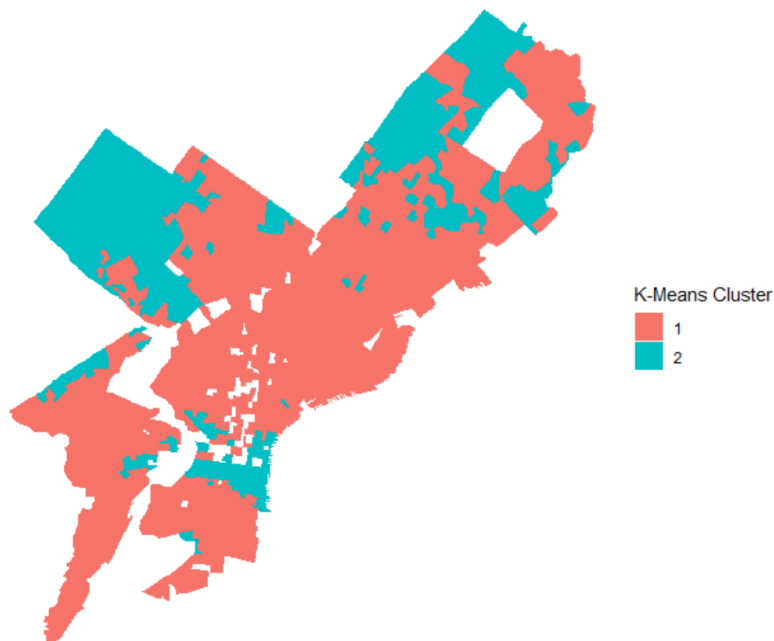


Figure 3: Spatial Distribution of the Clusters (k=2)

After creating the k-means cluster variable in R, we exported a CSV file containing POLY_ID and the cluster ID. I then joined this CSV file back to the Philadelphia block group shapefile using POLY_ID and mapped the spatial distribution of the clusters. This step follows the assignment requirement to export the cluster membership table and join it to RegressionData.shp by POLY_ID.

The map suggests that cluster membership is not randomly scattered across Philadelphia. Instead, block groups in the same cluster tend to appear near one another, indicating a visible spatial pattern in the cluster membership variable. Cluster 1 covers most of the city and appears especially common across large parts of North, West, and South Philadelphia. Cluster 2 is more spatially concentrated and appears in more limited areas, including parts of Center City, Northwest Philadelphia, and sections of Northeast Philadelphia. This pattern suggests that the higher-income, higher-education, lower-vacancy cluster is not evenly distributed across the city but is concentrated in specific parts of Philadelphia.

Looking at the map adds useful interpretation to the numerical cluster results. The aggregate table shows that Cluster 1 has lower income, lower home value, lower educational attainment, and higher vacancy, while Cluster 2 has higher income, higher home value, higher educational attainment, and lower vacancy. The map shows that these differences also have a spatial pattern. In other words, the clusters are not only groups formed from attribute similarity; they also show recognizable geographic patterns in Philadelphia's socioeconomic landscape.

The map also supports the descriptive names assigned to the clusters. Since Cluster 1 is widespread in areas with lower housing and income values, it will reasonably be described as "Lower-Advantage, Higher-Vacancy Neighborhoods." Cluster 2 is concentrated in areas with higher home values, higher incomes, and higher educational attainment, so it will be described as "Higher-Advantage, Lower-Vacancy Residential Areas."

Discussion

The k-means clustering results show that Philadelphia's block groups can be separated into two broad neighborhood types based on housing value, income, education, housing form, and vacancy. The first cluster includes most block groups and is characterized by lower median home values, lower household incomes, lower levels of bachelor's degree attainment, fewer single or detached-housing units, and higher vacancy rates. The second cluster is smaller but has much higher home values, higher incomes, higher educational attainment, more single or detached housing units, and lower vacancy rates.

The main pattern is that the two-cluster solution creates a broad socioeconomic divide across Philadelphia block groups. This result is not surprising because several of the input variables are closely related to neighborhood advantage and disadvantage. Areas with higher income also tend to have higher home values and higher educational attainment, while areas with lower income and lower home values tend to have higher vacancy rates. One important interpretation is that the final cluster solution appears to be driven more by overall

socioeconomic advantage than by a highly detailed set of neighborhood types. The spatial map reinforces this conclusion because the clusters form visible geographic patterns rather than appearing randomly scattered across the city.

One limitation of the result is that two clusters may oversimplify Philadelphia's neighborhood structure. A two-cluster solution is easy to interpret and is supported by the largest number of NbClust criteria, but it mainly divides the city into lower-advantage and higher-advantage areas. This may hide more nuanced neighborhood types, such as moderate-income areas, transitioning neighborhoods, or places with mixed housing and socioeconomic characteristics. Since the scree plot also suggests that two to four clusters could be reasonable, a three or four-cluster solution might provide a more detailed interpretation, even though the NbClust diagnostics most strongly support $k = 2$.

Overall, the k-means clustering analysis provides a useful summary of socioeconomic and housing variation across Philadelphia block groups. The final two-cluster solution is supported by the NbClust results and is easy to interpret substantively. However, the results should be understood as a broad typology rather than a complete description of Philadelphia's neighborhood structure. The analysis is most useful for identifying the city's largest socioeconomic and housing divide, while a larger number of clusters could be explored in future work to capture more moderate, transitional, or mixed neighborhood types.